Photoelectric Effect & Compton Effect

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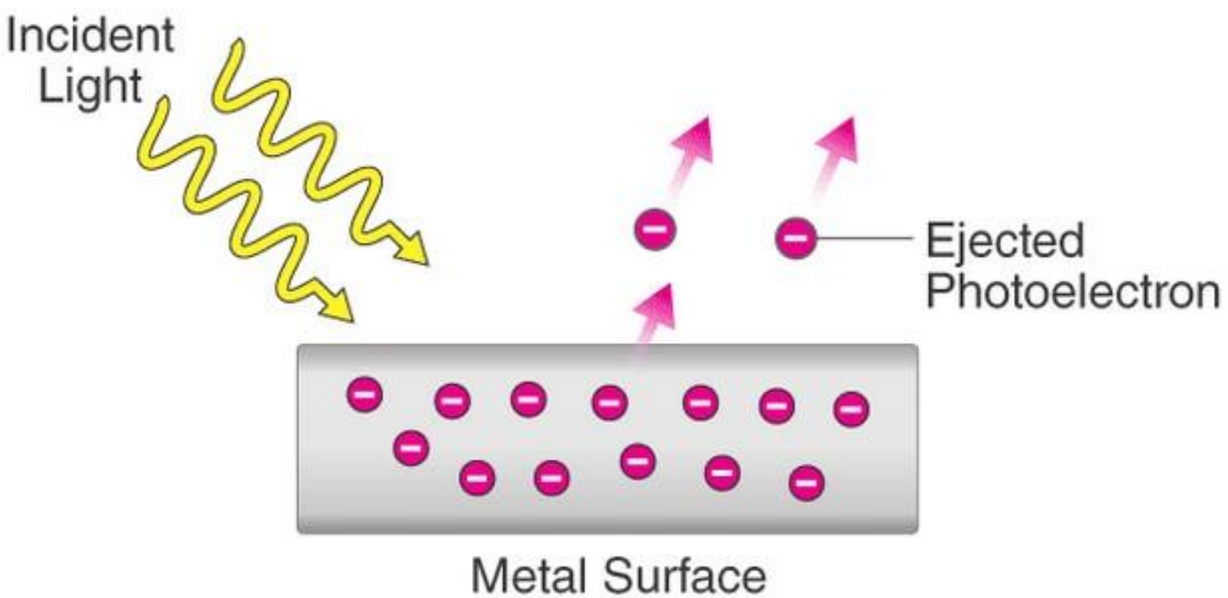
Whatever we will discuss

- Photoelectric effect
- Characteristics of photoelectric effect
- Einstein’s photoelectric equation
- Compton effect
- Related Mathematical Problems

Photoelectric Effect

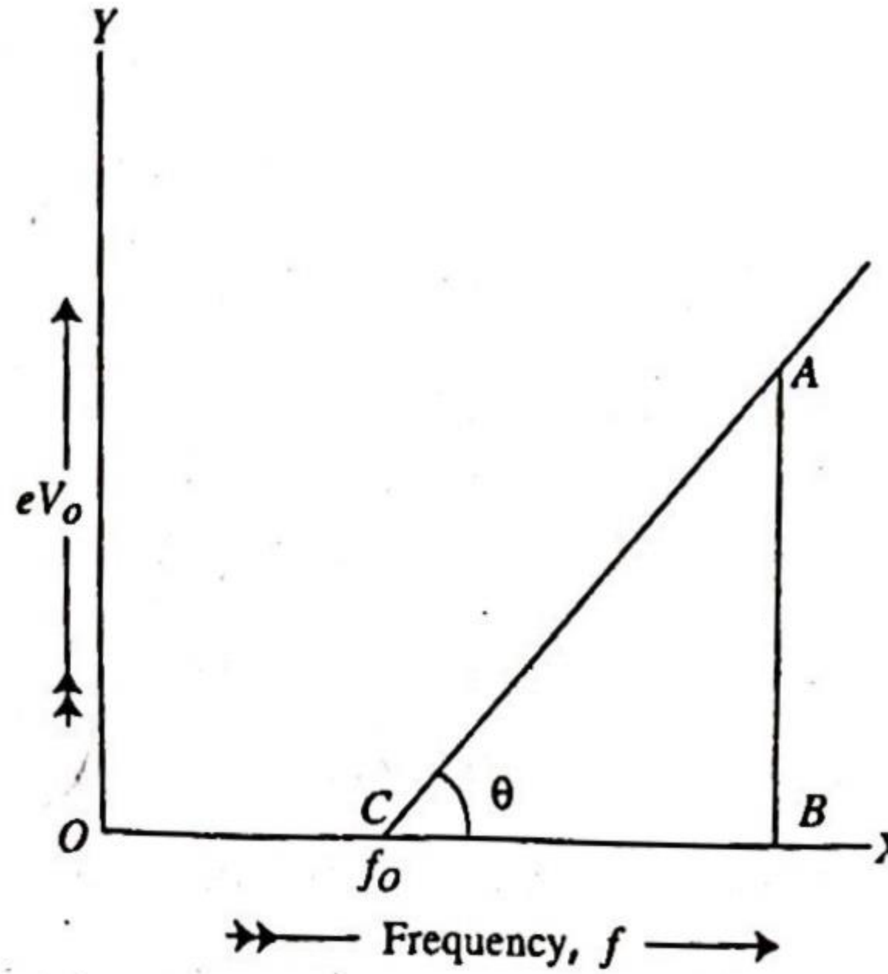
When light of appropriate frequency or wavelength is incident on a metal, then electrons are emitted from that metal. This process of emission of electrons is called **photoelectric emission** and the effect is called **photoelectric effect**. The electrons emitted due to the influence of light are called **photoelectrons** and current produced due to the emission of electrons is called **photoelectric current**.



Deduction of Einstein's equation: Though Einstein's equation for photoelectric effect cannot be deduced directly it can be established on the basis of experimental results. If the charge of the electron emitted from the metal plate with maximum velocity is e and stopping potential is V_o then the maximum energy of the photoelectron will be eV_o. If the maximum velocity of the photoelectron is v_m then maximum energy will be $\frac{1}{2} mv_m^2$.

$$\therefore \frac{1}{2} mv_m^2 = eV_o.$$

With the increase of frequency the energy of the electron eV_o increases. Now if we draw a graph taking frequency f along X-axis and eV_o along Y-axis then a straight line will be obtained which cuts the X-axis at point C [Fig. 12.7]. Here the point C indicates the threshold frequency f_o. A is any point on the straight line. From A perpendicular AB is drawn on the X-axis. Point B indicates any frequency f. If the straight line AC makes an angle θ with X-axis, then



$$\tan\theta = \frac{AB}{CB} = \frac{eV_o}{f - f_o}$$

Here, $\tan\theta$ = slope of straight line AC = constant. This constant is actually the Planck's constant h.

$$\therefore h = \frac{eV_o}{f - f_o}$$

or, $eV_o = hf - hf_o$

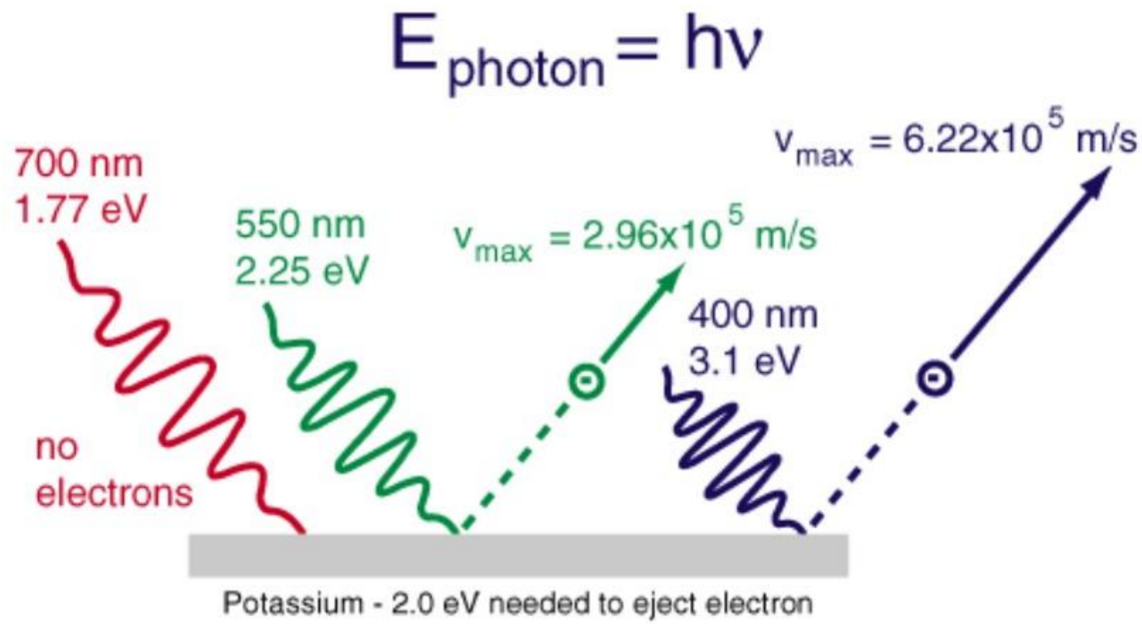
or, $\frac{1}{2} mv_m^2 = hf - hf_o$

But, $\frac{1}{2} mv_m^2 = K_{max}$ = maximum energy of photoelectron.

$$\therefore hf = K_{max} + hf_o$$

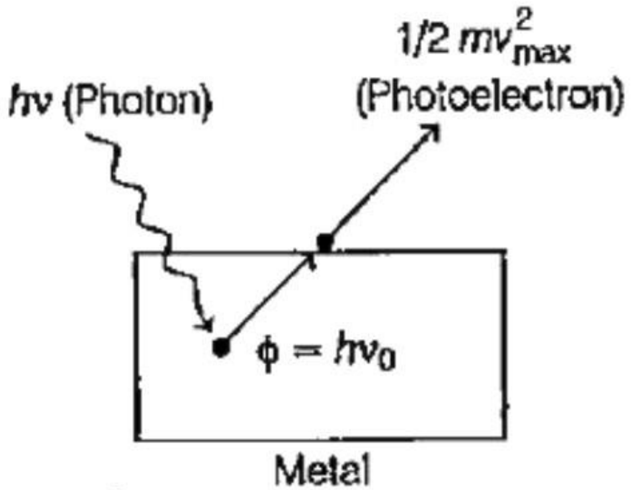
This equation is known as Einstein's equation of photoelectric effect.

Characteristics of Photoelectric Effect

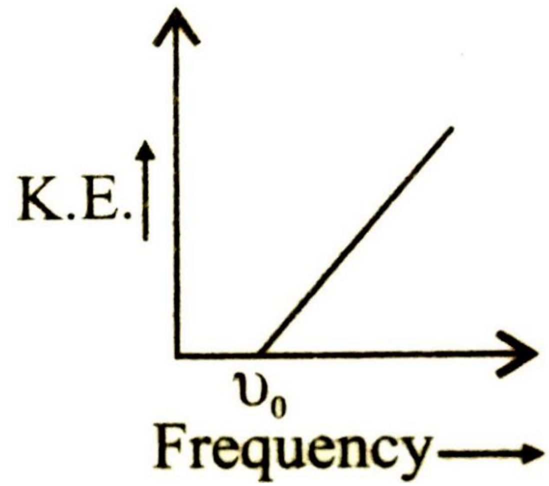


1. Photoelectric emission is an instantaneous phenomenon.
2. For the emission of electrons from each metal there is a minimum frequency of the incident light which is called the threshold frequency.
3. The maximum velocity (or, the kinetic energy) of photoelectrons is directly proportional to the frequency of the incident light.
4. The rate of emission of photoelectrons (or, the photoelectric current) is directly proportional to the intensity of the incident light.

Einstein's Photoelectric Equation



Emission of Photoelectron by a metal when a photon is absorbed by it



$$h\nu = \phi + KE_{max}$$

$$h\nu = h\nu_0 + \frac{1}{2} m v_m^2$$

$h\nu$ = Energy of each photon

ϕ = Work function

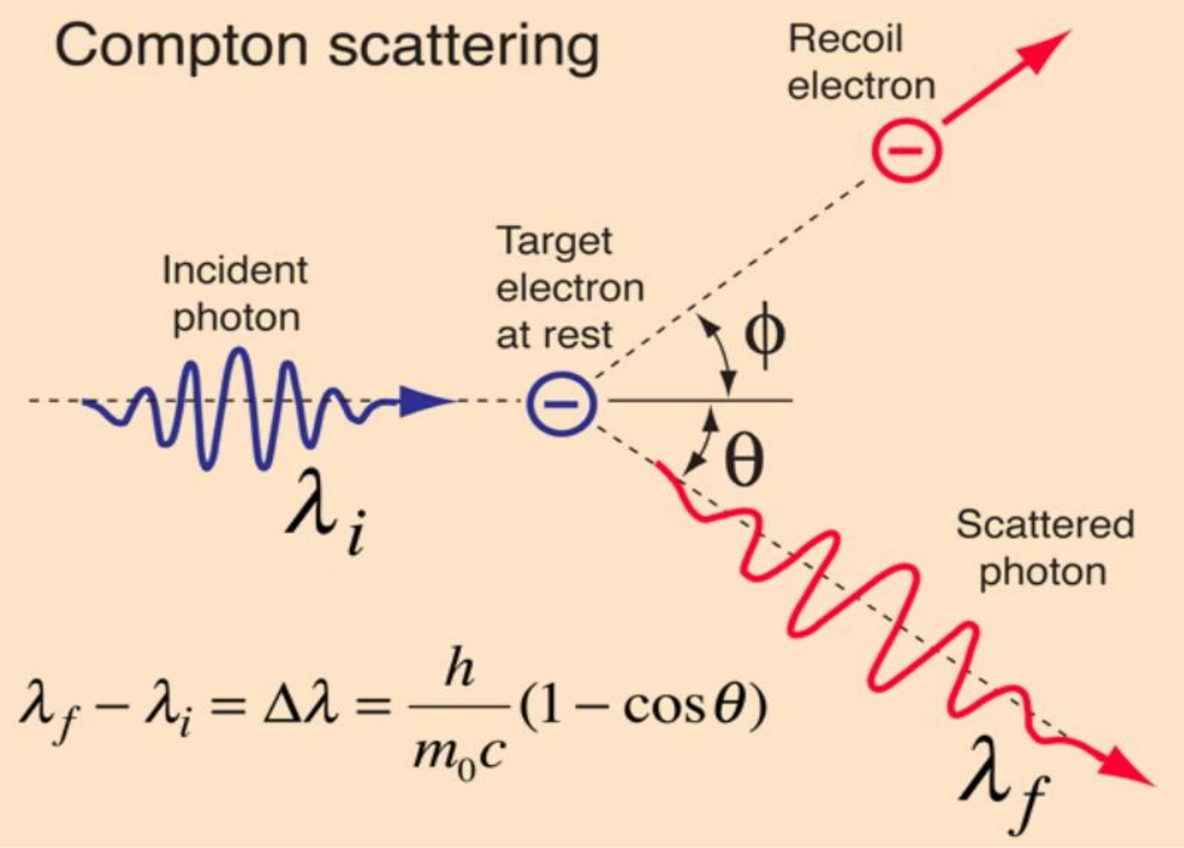
KE_{max} = Maximum kinetic energy

ν_0 = Threshold frequency

The amount of energy needed to emit electron having zero velocity from the surface of a metal is called the **work function** of that metal.

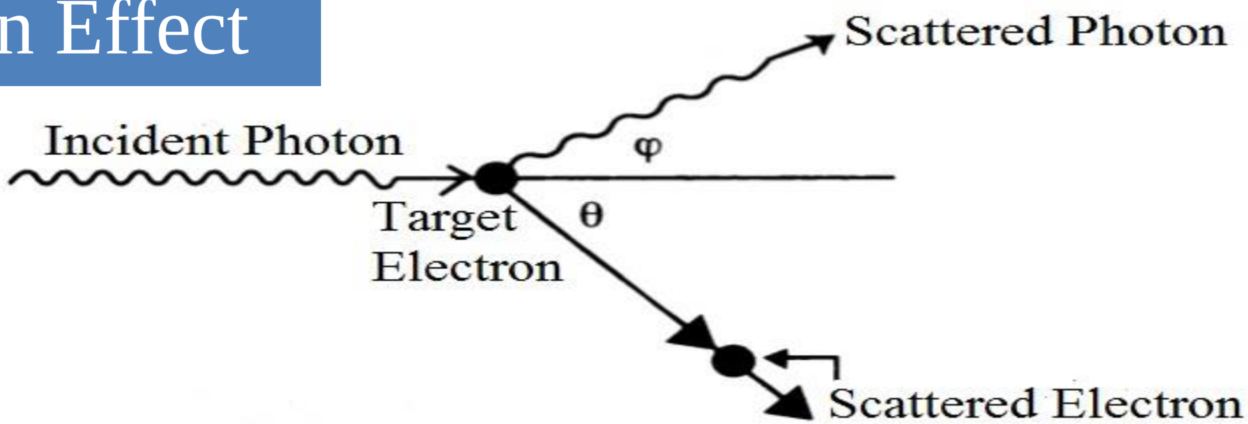
Compton Effect

The **Compton effect** is the enlargement of wavelengths of X-rays and other electromagnetic waves that have been scattered by electrons. It is a fundamental way in which radiating energy is absorbed by matter.



Compton effect (also called the Compton scattering) discovered by Arthur Holly Compton, is the scattering of a high frequency photon after an interaction with a charged particle, usually an electron. It results in a decrease in energy (increase in wavelength) of the photon (which may be an X-ray or gamma ray photon), called the **Compton effect**. Part of the energy of the photon is transferred to the recoiling particle.

Compton Effect



Important Question with Answer**: Find an expression for the difference of wavelength of the scattered photon and the wavelength of the incident photon due to compton effect.**

Let, due to the collision of a photon moving along X-axis with stationary electron, the photon and the electron have been scattered making angles φ and θ respectively with the direction of the incident photon

Let, the frequencies of the incident and scattered photon are f and f' respectively. So,

Energy of the incident photon, $E = hf$

$\therefore$ Momentum of the incident photon, $p = \dfrac{hf}{c}$

Energy of the scattered photon, $E' = hf'$

$\therefore$ Momentum of the scattered photon, $p' = \dfrac{hf'}{c}$

Since the electron was stationary before collision so its energy, $E_o = m_o c^2$ and momentum, $p_o = 0$. m_o is the rest mass of the electron.

If the velocity and mass of electron be v and m respectively after the collision then, its energy $= mc^2$ and momentum $= mv$.

Now according the law of conservation of energy,

$$hf + m_0c^2 = hf' + mc^2 \qquad \dots \qquad \dots \qquad \dots \qquad (8.69)$$

Along the incident photon that along X-axis the moments of the scattered photon and electron are respectively $\frac{hf'}{c}\cos\varphi$ and $mv\cos\theta$.

Along the perpendicular direction of the incident photon that is along Y-axis the momenta of the scattered photon and electron are respectively $\frac{hf'}{c}\sin\varphi$ and $-mv\sin\theta$.

So, according to the law of conservation of momentum for the component along X-axis,

$$\frac{hf}{c} + 0 = \frac{hf'}{c}\cos\varphi + mv\cos\theta. \qquad \dots \qquad \dots \qquad (8.70)$$

and for the component along Y-axis,

$$0 = \frac{hf'}{c}\sin\varphi - mv\sin\theta \qquad \dots \qquad .. \qquad (8.71)$$

The equation (8.70) can be written,

$$mvc\cos\theta = h(f - f'\cos\varphi) \qquad \dots \qquad \dots \qquad (8.72)$$

The equation (8.71) can be written,

$$mvc\sin\theta = hf'\sin\varphi \qquad \dots \qquad \dots \qquad (8.73)$$

Squaring and adding equations (8.72) and (8.73) we get

$$m^2v^2c^2 = h^2(f^2 - 2ff'\cos\varphi + f'^2\cos^2\varphi + f'^2\sin^2\varphi)$$

or, $m\,v^2c^2 = h^2(f^2 - 2ff'\cos\varphi + f'^2) \qquad \dots \qquad \dots \qquad (8.74)$

From equation 8.69 we get,

$$mc^2 = h(f - f') + m_oc^2$$

squaring both the sides,

or, $m^2c^4 = h^2(f^2 - f')^2 + 2h(f - f')m_oc^2 + m_o^2c^4$

or, $m^2c^4 = h^2(f^2 - 2ff' + f^{-2}) + 2h(f - f')m_oc^2 + m_o^2c^4 \qquad \dots \qquad (8.75)$

Subtracting equation (8.74) from (8.75) we get,

$$m^2c^2(c^2 - v^2) = -h^2ff'(1 - \cos\varphi) + 2h(f - f')m_oc^2 + m_o^2c^4$$

again, $m^2c^2 (c^2 - v^2) = \frac{m_o^2}{1 - v^2/c^2} c^2 (c^2 - v^2) = \frac{m_o^2 c^4}{c^2 - v^2} (c^2 - v^2) = m_o^2 c^4$

$\therefore m_o^2 c^4 = - h^2 ff'(1 - \cos\varphi) + 2h (f - f') m_o c^2 + m_o^2 c^4$

or, $o = - h^2 ff'(1 - \cos\varphi) + 2h (f - f') m_o^2 c^2$

or, $2h^2 \; ff'(1 - \cos\varphi) = 2h (f - f') m_o c^2$

or, $hff'(1 - \cos\varphi) = (f - f') m_o c^2$

or, $\frac{c(f - f')}{ff'} = \frac{h}{m_o c} (1 - \cos\varphi)$

or, $\left(\frac{c}{f'} - \frac{c}{f}\right) = \frac{h}{m_o c} (1 - \cos\varphi)$

But $\frac{c}{f} = \lambda$ = wave length of the incident photon

and $\frac{c}{f'} = \lambda'$ = wave length of the scattered photon

$$\therefore \lambda' - \lambda = \frac{h}{m_o c} (1 - \cos\varphi) \quad \ldots \quad \ldots \quad \ldots \quad (8.76)$$

$$\text{or, } \Delta\lambda = \frac{h}{m_o c} (1 - \cos\varphi) \quad \ldots \quad \ldots \quad \ldots \quad (8.77)$$

Here $\Delta\lambda$ is the difference of wavelengths of the scattered and incident radiations.

From equation (8.76) it is observed that $\lambda' > \lambda$ that is, the wavelength of the scattered photon is larger than the wave length of the incident photon.

Related Mathematical Problems

Photoelectric effect related mathematical problems

[Physics for Engineers (Part-2) by Dr. Gias Uddin Ahmad (pp. 437-456)]

Ex. 12.1, Ex. 12.2, Ex. 12.3, Ex. 12.4, Ex. 12.5, Ex. 12.6, Ex. 12.7, Ex. 12.9, Ex. 12.16, Ex. 12.18 & Ex. 12.19

Compton effect related mathematical problems

[Physics for Engineers (Part-2) by Dr. Gias Uddin Ahmad (pp. 520-525)]

Ex. 13.18, Ex. 13.19, Ex. 13.20 & Ex. 13.21

Thank You

Thanks for your attention